

Constipation and Laxatives

Constipation

- The term constipation can mean infrequent or hard stools, or it can refer to difficulty passing stools.
- In other words, constipation may involve pain during the passage of a bowel movement, the inability to pass a bowel movement after straining or pushing for longer than 10 minutes, or no bowel movements for more than 3 days.
- Reduced colonic motility and reduced frequency and bulk and increased hardness of feces surely do occur, mainly as a result of dehydration of material staying too long in the colon before expulsion.

Constipation

- The bulk, softness, and hydration of feces are very dependent on the fiber and water content of the diet; thus sufficient dietary fiber and water are mainstays in any regimen for the treatment of constipation.
- Since normal bowel patterns vary considerably from person to person, constipation is a relative term. Not everyone, for example, has a bowel movement every day.
- Constipation is not a disease but a symptom that may result from a broad variety of underlying causes, so may be congenital or result from endocrine disorders, diseases of the nervous system, and diseases of the large intestine or action of drug or toxins

Signs and Symptoms

- Infrequent, difficult passage of stools (fewer than three bowel movements a week)
- Sudden decrease in frequency of bowel movements
- Stools harder than normal
- Bowel still feels full after bowel movement
- Bloated sensation

Causes

1-low-fiber diet and habitual changes in food.

2-Inadequate intake of fluid each day

3- lack of physical activity.

4-Stress and travel.

5-delay in going to the bathroom when you have the urge to defecate.

6- diseases of the bowel (such as irritable bowel syndrome), pregnancy, certain medical conditions (like an underactive thyroid), depression, or medications may be the reason for your constipation

7-Uses of laxative in chronic periods lead to constipation.

Diagnosis

Your doctor will perform a physical examination, which may include a rectal exam and a pelvic exam if you are a woman. As part of the history, the doctor will ask a series of questions such as:

- How long have you had constipation?
- How many days between two bowel movements?
- Is your constipation worse when you are stressed?
- What is the color, shape, and consistency of the stools?
- Do you have any bleeding with bowel movements?
- Do you have any abdominal pain?
- What surgeries or injuries have you had?
- What medications do you take?
- Do you drink coffee or drink alcohol? Do you smoke?

Diagnosis

- The following tests may help diagnose the cause of constipation:
  - Blood tests such as a CBC, PT
  - Stool studies
  - X-rays of the abdomen
  - Barium enema (to look at the colon)
  - colonoscopy (an examination of the entire colon from the inside)

Treatment approach

- Chronic constipation can usually be prevented or treated with a combination of dietary changes, extra fluid intake, exercise, and, when necessary, short-term use of a laxative.
- Your health care provider may talk with you about proper bowel habits (consistent, unhurried elimination practices).
- He or she may have you use a laxative or stool softener over the short term or suggest a bulk-forming agent, such as psyllium, bran, or methylcellulose.
- Use laxative herbs with caution because, like medications, they may become less effective with habitual use.

Lifestyle

1-get enough fiber in their diet. Vegetables, fresh fruits (especially dried fruits) and whole wheat, bran, or oats are excellent sources of fiber.

To reap the benefits of fiber, you must drink plenty of fluids (especially water) to help pass the stool.

2-Regular exercise is also very important in establishing regular bowel movements.

3-Take time to eat, breathe slowly, and chew food thoroughly.

4-Eat smaller, more frequent meals and avoid overeating at one sitting.

5-Drink warm lemon water just before meals to stimulate digestion.

II- Constipation and Laxatives

Laxatives

- They are drugstaken orally to evacuate the bowel, and used in

1- Treatment of constipation.

2-Drug-food poisoning.

3-Before surgical procedure.

4-After anthelmintic drugs.

5-In radiological-examination as colonoscopy.

Classification of laxatives

Laxative effect and latency in usual clinical dosage		
Softening of feces, 1 to 3 days	***Soft or semifluid stool 6 to 8 hours***	***Water evacuation 1 to 3 hours***
Bulk-Forming Laxatives	**Stimulant Laxatives OR Antiabsorptive and Hydragogic Laxatives**	**Osmotic Laxatives**
Bran	**Anthraquinones derivatives**	**Magnesium sulfate**
Psyllium,	**Senna**	**Lactulose**
Linseed	**Rhubarb**	
Methylcellulose	**Cascara sagrada**	
	Frangula bark and Aloe Castor oil	

Stimulant Laxatives OR Antiabsorptive and Hydragogic Laxatives

- The drastic effect of anti-absorptive and hydragogue laxatives such as Senna, Rhubarb, Frangula bark and Aloe can be attributed to their anthranoid content.
- This group of substances is structurally related to anthracene, 1,8-dehydroxylation of the latter is necessary for development of the laxative effect.
- Some anthraquinones present in glycosides so by colonic microflora the active constituents will be liberated from their precursor glycosides.

Mechanism of action

- Anthranoids stimulate intestinal motility and inhibit the re-absorption of electrolytes and fluid from the colon, thereby increasing the volume of the intestinal contents (Faecal volume) and the intestinal filling pressure.
- This, in turn induces propulsive contractions, stimulates intestinal motility, and ultimately stimulate chloride secretion, resulting in the active secretion of water and electrolytes into the bowel.

Side effects of anthranoid drugs

- The monomeric aglycones especially emodin and aloe-emodin were shown to have genotoxic, carcinogenic and mutagenic effect.
- Long-term use of anthranoids may result in a reversible blackening of the colon (pseudomelanosis coli, PMC), which is due to incorporation of metabolites of the anthranoids and is thought to be associated with an increased risk of colon carcinoma, also loss of electrolytes.



Side effects of anthranoid drugs

•More immediate effects of anthranoid containing drugs are colic and griping pains due to increased spastic contractions of the smooth musculature of the GI tract, Aloe and Senna leaves are particularly prone to produce these.

•So these side effects must be compensated by addition of carminatives and antispasmodic drugs as Chamomile flowers, fennel and belladonna extract.

Side effects of anthranoid drugs

•Habituation may occurs from long-term use that lead to constipation.

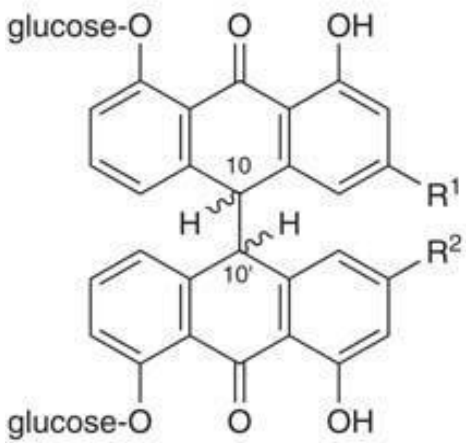
• Abnormal colour of the urine (yellowish brown turning red with increasing pH) during renal extraction of these compounds.

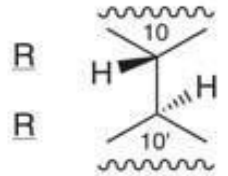
• Large doses may produce nephritis.

1- Senna

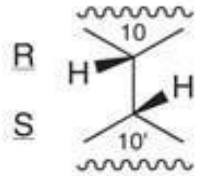
- **Constituents :**
- Dimeric anthraquinone glycosides Isomers of dianthrone type named sennosides A,B.C,D
- Very small amount of Aloe emodin and
- rhein in free state (aglycones)
- :

Hydroxyanthracene glycosides





dextro-rotation



optically inactive

	R^1	R^2	Isomeric form
sennoside A	COOH	COOH	(+), *trans*
sennoside B	COOH	COOH	*meso, cis*
sennoside C	COOH	CH_2OH	(+), *trans*
sennoside D	COOH	CH_2OH	*meso, cis*

Uses

- Laxative as it stimulates the muscular coat of the intestine and produce purgation which is not followed by constipation but should be used for short time and contraindicated for children under 12 years.
- Preparation of radiology or colonscopy
- Anti obesity (slimming preparations) but it has side effects as serious hypokalemia (depletion of potassium) and Hyperpigmentation of the colon.
- **Dosage and mode of Administration**

  - Warm water (ca 150 ml) is poured over half teaspoonful of Senna fruit then leave 10 minutes.
  - Cup of freshly prepared is drunk in the morning and/or at night before going to bed

Contra-indications, Warnings

- Itis recommended that senna should not be given to patients with

1-intestinal obstruction and stenosis

2- Atony

3-Inflammatory colon diseases (e.g. Crohn's disease, ulcerative colitis)

4-Appendicitis, with undiagnosed abdominal symptoms;

5-Severe dehydration states with water and electrolyte depletion.

- As with all laxatives, senna should not be given when any undiagnosed acute or persistent abdominal symptoms are present.
- Long-term use of laxatives should be avoided.
- Use for more than two weeks requires medical supervision.

Contra-indications, Warnings

- Chronic use may cause pigmentation of the colon (pseudomelanosis coli) which is harmless and reversible after drug discontinuation.
- Abuse, with diarrhoea and consequent fluid and electrolyte losses, may cause dependence, with possible need for increased dosages, disturbance of water and electrolyte (mainly hypokalaemia) balance, atonic colon with impaired function. Intake of anthranoid containing laxatives exceeding short-term use may result in an aggravation of constipation.
- Hypokalaemia can result in cardiac and neuromuscular dysfunction, especially if cardiac glycosides, diuretics or corticosteroids are also taken.
- Chronic use may result in albuminuria and haematuria.

B- Rhubarb

- **Constituents**

- The drug contains purgative anthranoids as rhein, rheum-emodin, chrysophanol as well as tannin.

- **Uses**

- Rhubarb act as purgative, dose deepened on severity of constipation, two, four, six or more of the rhubarb tablets must be taken with belladonna extract.

- It can be taken for extended periods of time, this is particularly important in the elderly patients with arteriosclerotic problems and susceptibility for gastrocardiac complaints.

- In small doses used as bitter stomachic and intestinal astringent.

- In larger doses it causes purgation, which is followed by an astringent effect due to presence of tannoid constituents

C- Aloe

- Barbados aloe is the condensed juice of the leaves of Aloe baradensis Miller, which contain at least 28% hydroxylanthracene derivatives with anhydrous barbaloin (aloin) as the principle base.
- Aloe is primary used as a laxative, but it also has choleretic action.
- Given in a high doses, aloe has drastic and abortifacient effects and, hence should not be used during pregnancy.

Pregnancy and lactation

- In view of the irritant and cathartic properties documented for aloes, its use is contra-indicated during pregnancy. Anthraquinones may be secreted into breast milk and, therefore, aloes should be avoided during lactation
- Aloes is reputed to be an abortifacient and to affect the menstrual cycle.

D- Frangula and Casacra barks

- They are used as a mild to moderate laxative, unlike Senn a and Aloe because its irritant effect in the intestines is not severe.
- Frangula bark can also be used to treat spastic constipation. With frangula bark, habituation does not occur as quickly as with senna, this makes frangula bark an ideal remedy for patients with habitual constipation who require a laxative tea for a while to restore normal bowel function.

Constituents

- **Frangula:**
- Glucofrangulin A and B which are diglucosides.
- **Cascara:**
- Cascarosides A, B, C, D, E and F which are stereoisomer sod aloin and derivatives, in addition to barbaloin, frangulin, palmidin A, B, ands C.

E- Castor oil

- The bean of the castor plant (*Ricinus commnuis*); contain two well-known ingredients; an extremely toxic protein ricin and oil composed chiefly of the triglycerides of ricinoleic acid.
- The usual cathartic dose in adults, 15 to 60 ml on an empty stomach, so castor oil should not be taken late in the day with the expectation of sleeping.

Metabolism, Catharsis and Adverse effects

- Externally castor oil used as emollient.
- Within the small pancreatic lipase hydrolyze the oil to glycerol and ricinoleic acid.
- Ricinoleate, like other anionic surfactants reduces net absorption of fluid and electrolytes and stimulates intestinal peristalsis.

Metabolism, Catharsis and Adverse effects

- Because ricinoleate acts in the small intestine, accumulation of fluid and evacuation take place rapidly and are relatively complete.
- The altered intestinal permeability caused by castor oil may reflect grosser morphological damage to the intestinal epithelium, the strong purgative action can cause colic as well as dehydration with the electrolytes imbalance.
- For these reasons and because of possible reduction of the absorption of nutrients, long-term use of castor oil must be avoided.
- The stimulant effects of this agent are reportedly sufficient to cause uterine contraction in pregnant women, who should, therefore, avoid using castor oil.

2- Bulk-Forming Laxatives and Lubricants

- These are bulking agents with a high percentage of fiber and are often rich in polysaccharides, which swell in the GI tract and increase in stool volume so intestinal peristalsis is stimulated due to the resulting stretch reflex.
- They influence the composition of food materials in GIT, especially via the colonic bacteria, which are thus provided with nutrients for proliferation. This turn influences the composition of the GIT flora and the composition of food in the tract (including an increase in gas, or flatus).
- Fiber-rich food is normally part of a health diet, but processed food and modern life styles have generally reduced fiber intake. Bulk-forming laxatives are generally not digested or absorbed in the GIT, but pass through it largely unchanged.

2- Bulk-Forming Laxatives and Lubricants

- Preparation of bulk-forming laxatives are always taken with the plenty of water at least one glass of water, they can be used to treat diarrhea if given with very little fluids; they then absorb the fluid from lumen and increase the consistency of the stool.
- Their advantages over stimulant laxatives are that they lack irritant effects, the onset of effect is not as rapid as that of herbal drastic laxatives, because it takes to begin working usually a few days. This makes them all the more suitable for long-term or even continuous use and consider save and free from side effects.

A- Linseed

- Linseed is a good source for fibers and high in roughage materials such as hemicellulose, cellulose, lignin and fatty oils, 52-78% linolenic acid ester.
- Linseed has both lubricant and bulk-forming effects and considered a good bulk laxative that increases the volume of stool mass, causing a stretch reflex that stimulate intestinal peristalsis.
- The fatty oil provides additional lubrication for the larger stool mass. Hence, the two actions of linseed complement each other beautifully, there is no danger of side effects, even the trace amounts of hydrocyanic acid and its derivatives do not have injurious effects.

B- Plantago psyllium, Psyllium plantago **(Ispaghula)**

Plantago psyllium (Plantaginaceae); and *Psyllium plantago ovata* (Ispaghula), both plants contains **a higher percentage of mucilage so have a high swelling factor**. Ispaghula seed work by stimulating the intestinal peristalsis as mentioned before, also used in diarrhea due to fluid binding??.

Plantago is very useful and commonly used emollients and bulk laxatives which help in maintaining a regular bowel movement.

C- Wheat bran

- Bran is less useful as a laxative except when taken as a natural part of the diet e.g. in breakfast cereal) since it contain phytic acid, which in high concentrations can complex with and therefore reduce the bioavailability of vitamins and minerals taken at the same time. However, in some patients wheat bran is more effective than other swelling agents, and taken with water.

3- Osmotic Laxatives

- Another useful approach to constipation is the use of osmotic laxative such as lactulose or lactose, two dimeric sugars derived from milk.
- Lactose is split in the GI tract into glucose and galactose. Adults generally do not resorb galactose well, consequently the bacteria of the colon metabolize this sugar.
- The resulting acids, including lactic acid and acetic acid, have an osmotic effect and the bacteria in the colon multiply more rapidly.
- This results in an increase in the amount of faeces with a subsequent increase in GIT peristalsis.

Other Considerations

•**Pregnancy**

•Constipation is common in pregnancy and is usually relieved by changing your diet and drinking more water.

• If you are pregnant, DO NOT take herbs that are stimulating to the digestive tract since they can induce contractions.

• In fact, you should not use any laxative herbs during pregnancy without a provider's supervision.

•**Prognosis and Complications**

•The passage of large, wide stools may tear the mucosal membrane of the anus, especially in children.

•This can cause bleeding and the possibility of an anal fissure.

Irritable Bowel Syndrome

•Irritable bowel syndrome (IBS) occurs when muscles in your intestines contract faster or slower than normal.

•This causes pain, cramping, gassiness, sudden bouts of diarrhea, and constipation

•. In the past IBS has been called mucous colitis,

•spastic colon, and irritable colon, but these terms are misleading because the problem is not limited to the colon

Two types of IBS

1-Spastic colon IBS:

You experience constipation, diarrhea, or both, and you often have pain after eating.

2-Painless diarrhea IBS:

Involves the sudden onset of diarrhea during or after meals, or upon walking.

- Between 10 and 20 percent of the population has IBS at some time.
- The syndrome often starts in young adults.
- It affects three times as many women as men and is often associated with stress.

Signs and Symptoms

- Cramping pain in your lower abdomen
- Bloating and gassiness
- Change in bowel habits.
- Diarrhea or constipation, or both alternately
- Immediate need to evacuate your bowels when you wake up or during or after meals
- Relief of pain after bowel evacuation.
- Feeling of incomplete emptying after bowel evacuation.
- Mucus in your stool.
- Depression and anxiety.
- Headache and fatigue.
- Nausea.

RISK FACTORS

- **Age**:
- IBS typically begins around age 20 and rarely appears after age 50.
- **Gender**:
- Two to three times as many women as men have the condition.
- **Genetics**:
- Many with IBS have a family member with the disorder, suggesting a possible genetic cause.
- There are also a variety of factors that can trigger an episode such as emotional stress.
- Many people find that their symptoms worsen when they eat certain foods, such as milk, caffeine, wheat, alcohol, artificial sweeteners, or food preservatives. Other illnesses, such as an infection, can trigger IBS.

What Causes It?

The underlying cause remains unknown.

But the syndrome has no relation to actual disease, and it does not lead to other diseases.

Several factors are thought to contribute to IBS symptoms, including:

- Emotional stress
- Eating habits
- Elevated hormone levels (in women)

Treatment Options

- Try to avoid stressful situations or foods that have triggered IBS in the past.
- Monthly hormonal changes and some drugs can affect your condition.
- Establishing regular bowel habits can be helpful.

Drug Therapies

1-Anticholinergic agents

reduce the pain from bowel spasm

2-Antiflatulents

reduce gas

3-Antidiarrheal medication

such as Loperamide (Imodium)

4-Bulk-producing agents

such as psyllium (Metamucil) and other fiber supplements can help with diarrhea or constipation

CAM

1- Nutrition

- Removal of known food allergens or irritants is important.
- The most common food allergens are wheat, corn, peanuts, citrus, eggs, fish, and tomatoes.
- Eliminate all suspected allergens from the diet for two weeks.
- Add back one food every three days and wait for reaction.

CAM

.If you suffer from gassiness, eliminate beans, cabbages, and other "gassy" vegetables from your diet, as well as apple juice, grape juice, bananas, nuts, and raisins.

.Fiber supplementation can help reduce pain, cramping. Supplements include psyllium, slippery elm, and marshmallow root.

.• Digestive enzymes taken 20 minutes before meals can help enhance digestion and normalize bowel function.

CAM

- One teaspoon of raw bran with each meal, supplemented by extra fluids, provides fiber reliably.
- Pro-flora supplements such as acidophilus and lactobacillus species taken two to three times per day can help to rebalance normal bowel bacteria and reduce gas and bloating.
- Magnesium (200 mg two to three times per day) and B-complex (50 to 100 mg per day) with extra B5 (pantothenic acid; 100 mg per day) may help reduce the effects of stress.
- Low-fat diets may relieve abdominal pain following meals.

2- Herbs

1.Enteric-coated peppermint oil:

one to two capsules (0.2 ml peppermint oil per capsule) three times a day after meals

1.A tea of fennel or ginger root

taken after meals promotes good digestion

3.A tincture of equal parts

1.of the following before meals (30 drops three times per day): valerian, passion flower Ã Anise extract, and milk thistle (Silybum marianum)

Herbal remedies

- **Mentha**
- contain volatile oil the main components being menthol, menthone, menthylacetate, and menthofuran
- **Pharmacological effects and clinical efficacy**

The antispasmodic effects of peppermint oil have been well established and due to Ca+ antagonistic effects on the smooth muscle of the GIT, beside carminative action

B- Fruits of the Apiaceae (Umbelliferous fruit).

- The fruits of several members of family Umbelliferae; are used as carminatives as they are rich in essential oil and have an antispasmodic effects.
- **Caraway (*Carum carvi* L.)**
- The drug contain essential oil as (+)- carvone responsible for **the spasmolytic action**.
- **Pharmacological effects and clinical efficacy**
- The aqueous extract, and even more so the essential oil act as a spasmolytic

The fruit are used in cases of dyspepsia, minor GIT cramps and flatulence, in addition to caraway used in infant gripe water.

B- Fruits of the Apiaceae (Umbelliferous fruit).

- **Fennel**
- fennel containing 2-6% Volatile oil (60% *trans*-anethole and 15% fenchone.
- **Pharmacological effects and clinical efficacy**

Fennel is used as a carminative, for indigestion and colic in children.

It is considered to be a very safe drug and is widely used as a health-food supplement as well as a spice.

- **Other umbelliferous fruits used:**
- Anise, Coriander and star anise.

3- Physical Medicine

1.Electric heating pads) hot water bottles, and long hot baths can relieve painful spasms and cramping in the abdomen.

2.Regular exercise, such as walking, can reduce stress and encourage bowel movements if you are constipated.

3.Castor oil pack.

Apply oil directly to skin, cover with a clean soft cloth and plastic wrap. Place a heat source over the pack and let sit for 30 to 60 minutes.

Dietary Recommendations

Foods to include:

•probiotics, such as live yogurt, can help restore normal gut flora, aid digestion, and reduce symptoms of IBS.

•Drink lots of fluids, such as water, herbal tea, and vegetable and fruit juices, throughout the day.

•Eat small, frequent meals (four or five meals daily), chew food thoroughly, and eat slowly. This will help regulate bowel function and improve digestion.

•Fibres can help regulate bowel function and reduce constipation, but it can also cause gas and cramping. Add high-fibre foods to the diet gradually to minimize gas and bloating and drink plenty of water. Recommended fibre-rich foods include oats, brown rice, milled flaxseed, psyllium, and vegetables.

Foods to avoid

- Alcohol, caffeinated beverages, and spicy foods stimulate the intestines and can make diarrhea and pain worse.
- Artificial sweeteners, such as sorbitol or mannitol, cause reactions (gas and diarrhea) in some people. Sorbitol and mannitol are found mainly in diabetic and low-carbohydrate foods and snacks.
- • Carbonated drinks (soda water) can produce gas and should be avoided.
- • Chewing gum causes ingestion of swallowed air, which contributes to intestinal gas.
- • Concentrated amounts of fructose (fruit sugar) in dried fruit and fruit juice can ferment in the colon, causing gas and bloating.